Seaborn & Pandas Data Exploration Output

Exploratory Data Analysis

Extracting critical guest behaviors, seasonal dependencies, and metric correlations to construct highly accurate predictive models.

01 / Dataset Distributions & Outliers

Examining general feature metrics, numerical distributions, correlation strengths, and outlier behaviors across variables.

Numerical Features Distribution

Initial Distribution Review

Analyzing individual numerical attributes highlights underlying feature skewness and scale disparities:

- **Average Room Price:** Shows a near-normal distribution peaking between 80 to 120, with high-end outliers.
- **Special Requests:** Highly right-skewed; most guests make zero or one request, indicating high-intent customer sub-segments.
- **Preparation Insights:** Direct visual feedback justifies mapping discrete and continuous columns differently prior to model input.

Dataset-wide Attribute Distributions

Screenshot 2026-07-16 211801.jpg

Range Analysis & Outliers

Seaborn Scale & Variance Boxplot

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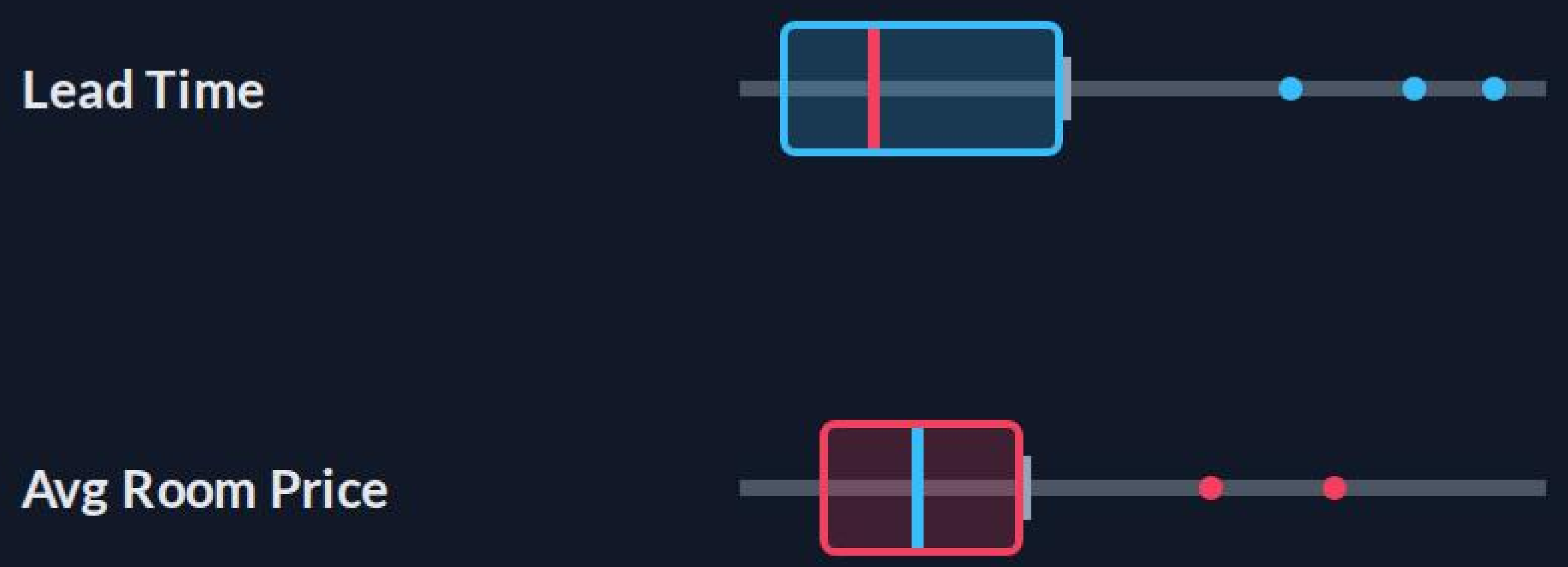

Evaluating Data Variances

Inspecting raw boxplots provides the baseline statistical justification for applying specific scaling methods:

- **Lead Time Variance:** Values extend far past 300+ days, showcasing heavily skewed planning structures.
- $ **Average Daily Rate:** Isolated outlier points exceed 500 units, indicating holiday inflation or luxury suites.
- **Transformation Strategy:** Robust scaling or log transformations (such as log1p) are necessary to normalize tree splits.

Multidimensional Correlation Matrix

Linear Feature Interactions

Analyzing numerical correlations reveals key dependencies that guide our predictive model selection:

- **Adults & Room Price:** Positive correlation (0.33) highlights that room prices scale strongly with double-occupancy rooms.
- **Previous Stay Dynamics:** Repeated guests have moderate positive correlations (0.38 to 0.56) with previous cancellations and bookings, revealing loyal segment patterns.
- **Sparse Features:** No strong multicollinearity exist overall, making linear or tree-based classifiers highly robust.

Feature Correlation Heatmap Matrix

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Adults	1.0	-0.05	0.16	0.33	0.16
Children	-0.05	1.0	0.01	0.35	0.09
Lead Time	0.16	0.01	1.0	-0.04	0.00
Room Price	0.33	0.35	-0.04	1.0	0.17
Requests	0.16	0.09	0.00	0.17	1.0

02 / Key Predictors of Cancellations

Inspecting lead times, average room prices, requested services, and meal selections to pinpoint booking commitment indicators.

Impact of Lead Time & Room Price

Boxplots split by Cancellation Status

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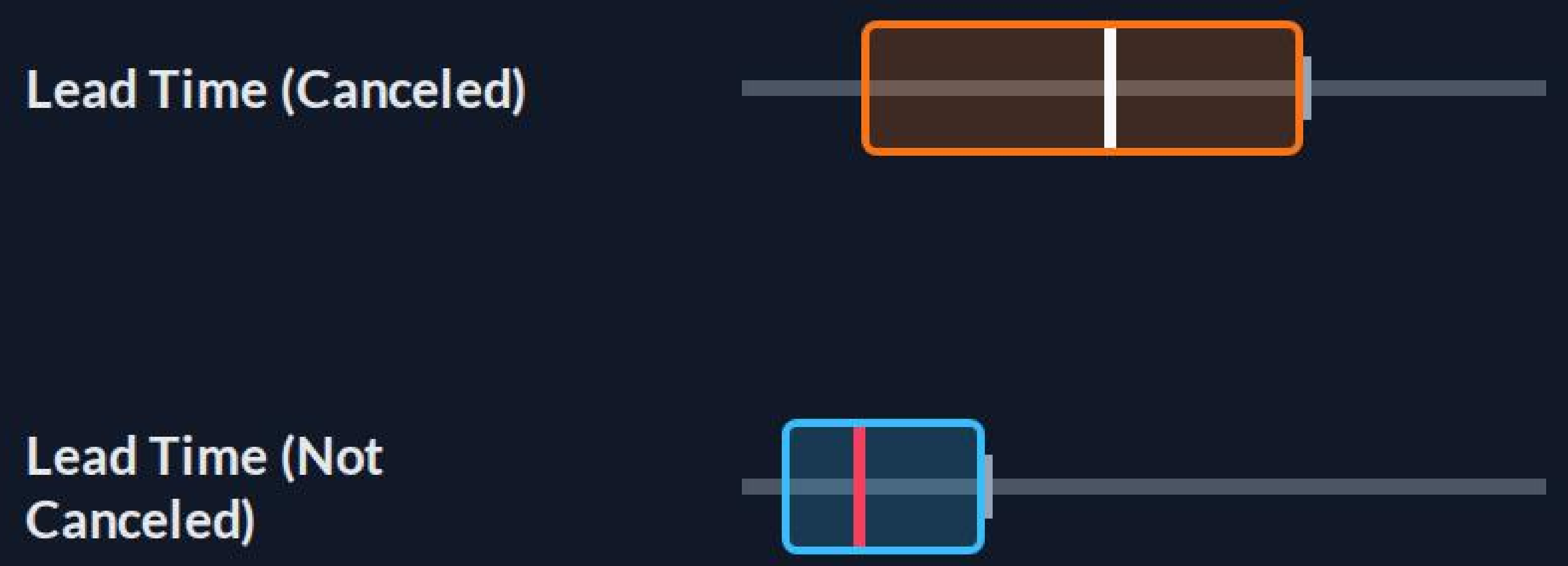

Behavioral Divergence

Analyzing boxplots split by checkout outcomes highlights strong predictive signals:

- **Lead Time Split:** Canceled bookings feature a significantly higher median lead time (~100 days) compared to completed stays (~35 days). Long planning windows raise cancellation rates.
- **Average Room Price:** Canceled stays feature higher median price ranges, proving that budget constraints influence the decision to drop a reservation.

Meal Plans & Special Requests

Intent & Customization Factors

The correlation between customized services and actual check-ins represents a massive business opportunity:

- **Special Requests:** Guests making 1+ special requests cancel at exponentially lower frequencies. Seeking tailored perks represents a reliable commitment signal.
- **Meal Choices:** "Meal Plan 1" is the undisputed dominant choice, yet its high cancellations correlate closely with standard retail OTA rates.

Special Requests Countplot Analysis

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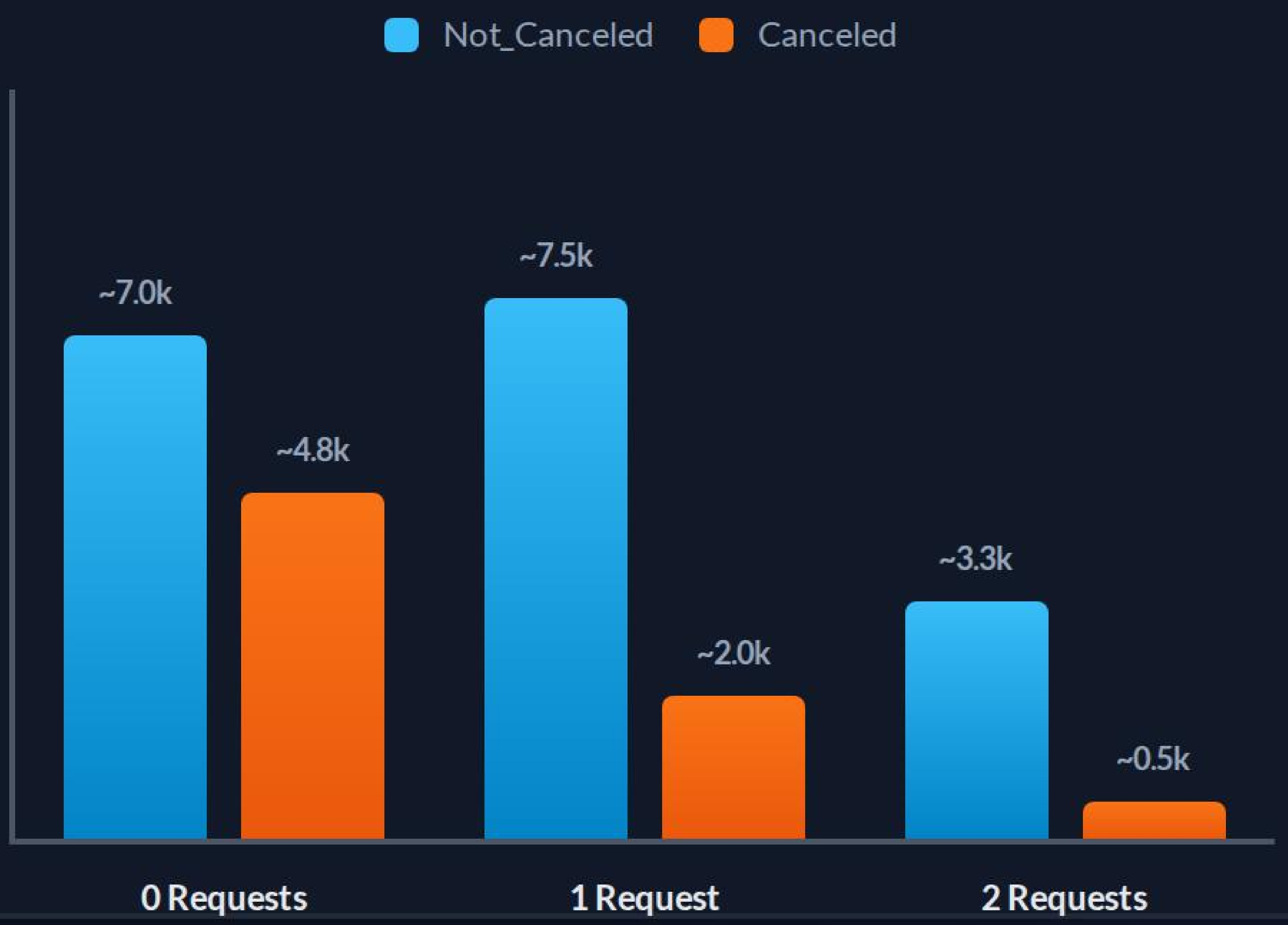

03 / Guest Segments & Seasonality

Deciphering family demographics, recurring loyal cohorts, and monthly revenue demand fluctuations.

Guest Segment Demographics

Adult Counts Segmented by Status

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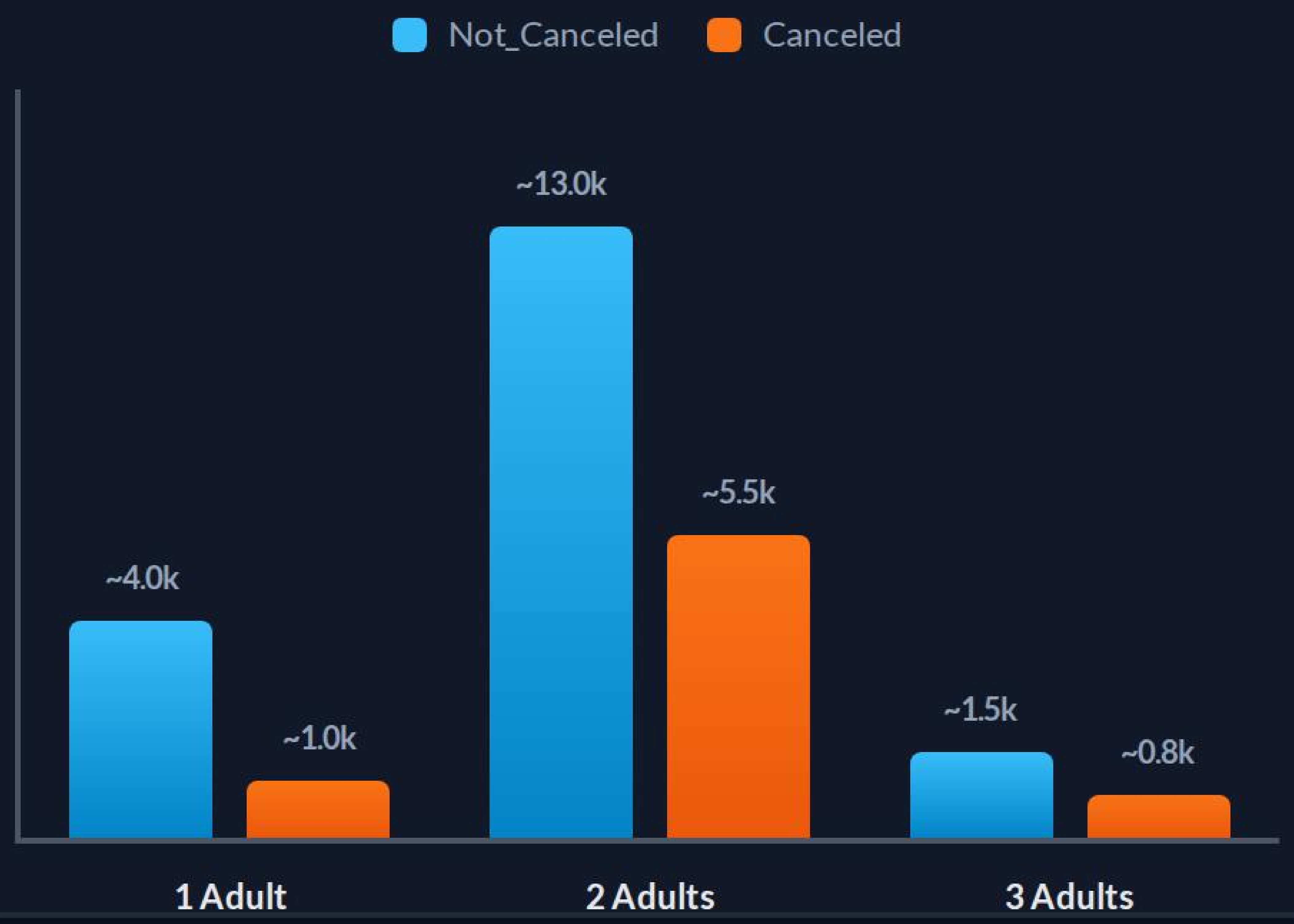

Occupant Profiles

Identifying our customer segments reveals the differences between single, double, and family accommodations:

- **Adult Dominance:** Stays are predominantly double-occupancy (2 adults), which displays the largest share of overall volume and cancellations alike.
- **Children Counts:** The vast majority of reservations include zero children. However, family bookings (with 1 or 2 children) exhibit higher fulfillment rates once requested.

Loyalty & Utility Indicators

High-Intent Commitment Signals

Analyzing behavioral choices reveals two highly committed user groups with minimal cancellation rates:

- **Repeated Guests:** Returning customers rarely cancel, representing the most stable source of recurring hotel revenue.
- **Parking Spaces:** Guests requesting parking spaces display virtually zero cancellations. This reflects high-intent road travelers with localized plans.

Parking Demands vs. Checkout Outcomes

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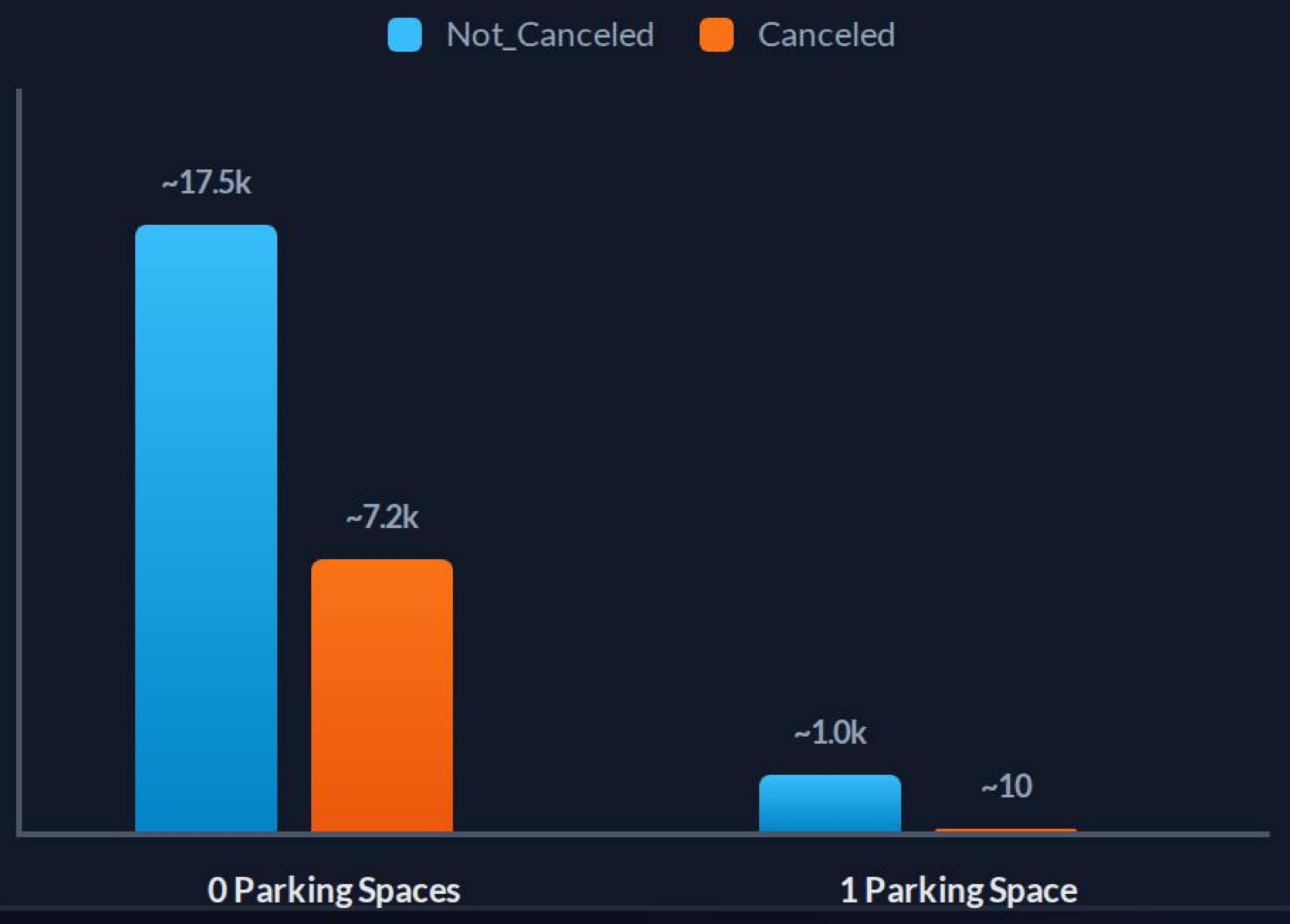

Seasonality & Pricing Models

Monthly Average Room Rate (ADR) Trends

Screenshot 2026-07-16 212030.jpg

Seasonal Revenue Fluctuations

Monthly trends reveal clear seasonal pricing strategies and demand shifts:

- **Summer Surcharges:** Room rates peak during summer (Months 6-8) at ~120 units, resulting in higher overall cancellation volumes.
- **Winter Low-Season:** Winter bookings drop significantly in pricing (75 units) and record minimal cancellations, representing low-risk revenue.

Statistical Divergence Summary

Booking Status	Lead Time (Avg)	Avg Price / Room	Special Requests (Avg)	Adults (Avg)	Car Parking Required
Canceled	104.36 Days	115.34 Units	0.41 Requests	1.97 Guests	1.4% of stays
Not Canceled	50.33 Days	101.98 Units	0.87 Requests	1.85 Guests	5.3% of stays

Data Summary (Screenshot 2026-07-16 212124.jpg): Grouping data by outcome confirms that canceled bookings have double the lead time, higher room prices, and fewer than half the average special requests compared to completed stays.

Questions & Answers

Applying exploratory insights to fuel optimal predictive models.

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